

DBCLS BioHackathon 2026 report: A knowledge base for the Marchantia genome database MarpolBase, and AI-assisted tools for DDBJ submission

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Introduction

Our group worked on two loosely coupled goals during the DBCLS BioHackathon 2026.

The first was to turn **MarpolBase** (Tanizawa et al., [2025](#)), the genome database of the liverwort *Marchantia polymorpha* (Bowman et al., [2017](#)), into a knowledge base: to publish its gene, expression and literature data as RDF, to expose them through a SPARQL endpoint, and to make them reachable from AI agents through a Model Context Protocol (MCP) server.

The second was to lower the cost of **depositing data in DDBJ**. Preparing a DDBJ submission file is still largely manual work, and the two side projects reported here — DFAST-Agent and MARKit — attack that problem from two directions: an agent that assembles submission files through dialogue with the user, and a tool that builds and checks marker-sequence submissions mechanically.

Table 1 lists what was produced.

Table 1: Resources and tools developed or extended during BioHackathon 2026.

Name	Description and URL
MarpolBase RDF	RDF distribution of the MarpolBase gene, expression and literature data. https://marchantia.info/rdf/
MarpolBase SPARQL endpoint	Public endpoint, registered with TogoMCP and RDF Portal. https://marchantia.info/sparql
MarpolBase MCP server	14 tools giving agents typed access to genes, expression, co-expression and literature. https://marchantia.info/mcp
DFAST-Agent	Experimental prototype of an agent that assembles DDBJ submission files, covering prokaryotic genomes (via the DFAST API) and eukaryotic genomes (from GFF). A web interface backed by a local LLM was also prototyped. Not released.
MARKit	Marker Annotation and Registration Kit, distributed as the container <code>nigyta/markit:latest</code> . https://ggs-staging.ddbj.nig.ac.jp/tools/markit

MarpolBase RDF and MCP server

What was published

The gene, expression and literature data served by MarpolBase were converted to RDF and released. The SPARQL endpoint was registered with both TogoMCP and RDF Portal, so

MarpolBase can now be reached from the cross-database query tooling maintained in Japan. On top of the endpoint we built an MCP server with 14 tools, which lets an LLM agent query the resource without writing SPARQL by hand.

The tools fall into five groups: gene lookup and search, expression profiles and condition comparison, co-expression networks, curated gene–literature links, and controlled-vocabulary resolution. A `run_sparql` tool and a `describe_schema` tool are also provided so that an agent can fall back to raw SPARQL when the typed tools are not enough.

Use cases

We exercised the server with two analyses driven entirely from natural-language prompts.

In the first (Figure 1), the prompt “*tell me the conditions in which MpBNB is highly expressed*” was answered by retrieving the mean TPM of MpBNB across all 161 conditions in MarpolBase, together with the condition attributes (tissue, accession, stage, sex, mutant, treatment, infection). The gene reaches 27.8 TPM in the antheridium and stays near zero in thallus, gemma and spore, which the agent could report because the expression values carry condition metadata rather than bare sample identifiers.

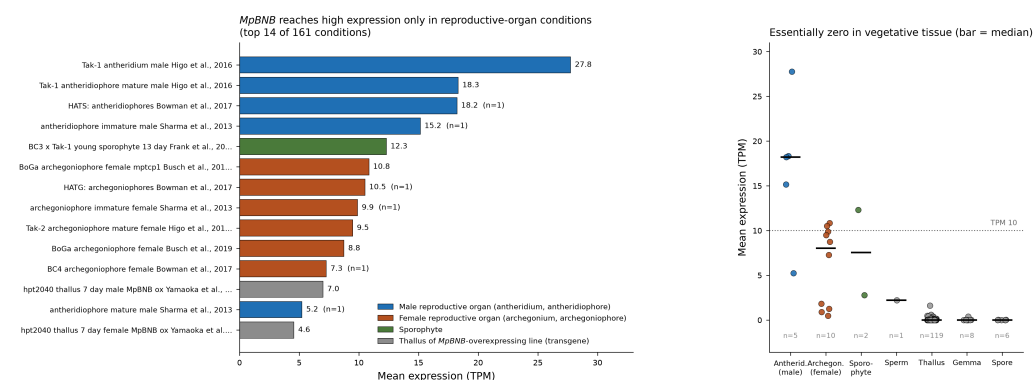
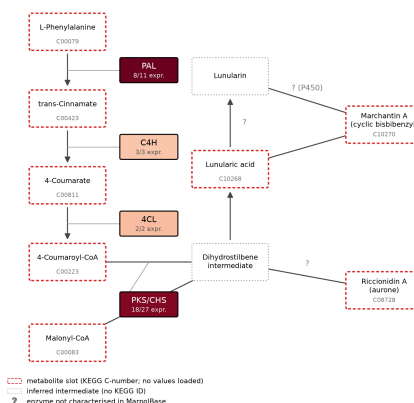


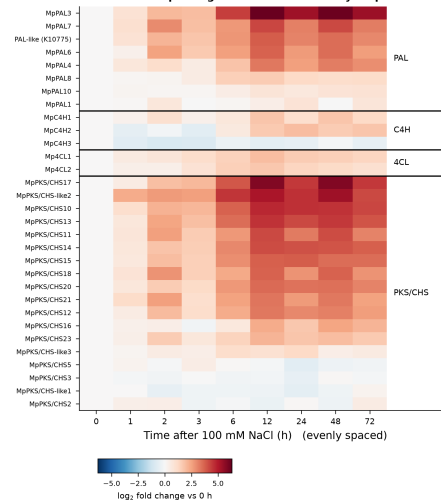
Figure 1: Expression of *MpBNB* across the 161 conditions in MarpolBase, retrieved through the MCP server from a single natural-language prompt. Left: top 14 conditions by mean TPM, coloured by organ type. Right: the same values grouped by tissue.

In the second (Figure 2), we followed the bisbibenzyl biosynthesis route under salt stress. Expression values came from MarpolBase; the pathway context — compound identifiers, reactions and enzyme references — came from external resources reached through TogoMCP. The committing PKS/CHS step and the shared phenylpropanoid entry are both salt-induced, and the induction is paralog-selective within every step.

a Bisbibenzyl route: the committing PKS/CHS step is salt-induced, like the shared phenylpropanoid entry



b Induction is paralog-selective within every step



Panel a: enzyme boxes coloured by the strongest induction among that step's expressed paralogs; 'N exp' = paralogs above the expression floor / all paralogs annotated in MarpolBase. Panel b: rows are individual genes (log2 fold change vs 0 h), grouped by step and sorted by 12 h response. 31 of 43 genes shown; 12 omitted — 11 below the 1 TPM floor (a fold change from a -0.01 TPM baseline is not interpretable) and MgPAL5 (MgPAL5 [4170], whose profile MarpolBase returned an error for and was therefore not assessed, not treated as absent). Steps from lunularic acid onward have no characterized enzymes in MarpolBase and are left uncoloured. Reaction topology is a literature schematic, NOT from KEGG — lunularic acid (C00088), marchantin A (C0070) and riccionidin A (C00728) exist in KEGG COMPOUND but are placed on no KEGG pathway map. Expression: Tai-1 thallus 7 d, Wu et al., 2021 (n_runs 3-7 per time point).

Figure 2: The bisbibenzyl route under salt stress. (a) Pathway schematic with enzyme steps coloured by the strongest induction among the expressed paralogs of that step. (b) log2 fold change of individual genes over a 100 mM NaCl time course, grouped by step.

What MarpolBase and TogoMCP each contribute

Working through these two analyses made the division of labour between a species-specific resource and a cross-database hub concrete (Table 2).

Only MarpolBase can supply measured expression across its 161 conditions, the co-expression network (PCC / HRR / MR), DESeq2 differential expression between two conditions, gene-literature links curated with evidence and role, and controlled-vocabulary resolution with PECO and PO identifiers. It also carries its own KEGG Orthology assignments — `mpo:hasKEGG` for 6,563 genes. This last point mattered in practice: *M. polymorpha* is absent from KEGG GENES, and the enzyme nodes in Figure 2 could be coloured **only** because MarpolBase assigns KO numbers itself.

Only TogoMCP can convert identifiers across 119 databases (TogoID) (Ikeda et al., 2022), search the species-agnostic references (UniProt, PDB, Reactome, Rhea, ChEMBL, MeSH), reach the NCBI E-utilities for SRA, GEO and PubMed, and dispatch SPARQL to a choice of endpoints.

Table 2: Where the two resources overlap, and how the overlap differs in kind.

Capability	MarpolBase	TogoMCP
SPARQL	One own graph (GRAPH must be given explicitly)	A choice of endpoints
Literature	<i>Marchantia</i> papers linked to genes, with evidence and role	Generic PubMed / MeSH search
Compounds	Not covered	ChEBI / Rhea / PubChem identifier conversion
Gene expression	Measured values for 161 conditions	Not held

The two do not meet at the gene identifier

The most useful negative result of the project: **TogoID holds no *M. polymorpha* gene dataset, and MarpolBase's `get_gene` does not return UniProt accessions.** There is therefore no identifier-conversion path that carries Mp3g23300 directly into the TogoMCP side. We also confirmed that pfam → uniprot does not exist as a TogoID route (404), while rhea/chebi, chebi/pubchem_compound, uniprot/rhea, uniprot/pdb and go/uniprot do (in both directions).

The two resources meet instead at the level of **shared vocabularies**:

1. KO numbers — the junction that made the two-layer map in Figure 2 possible.
2. GO and Pfam — assigned by MarpolBase, dereferenceable on the TogoMCP side.
3. Compound identifiers (ChEBI / PubChem) — the junction for metabolome data.

Going through UniProt instead means searching by protein name and organism, which for paralog families such as MpPAL1–10 is many-to-many and therefore unreliable.

The practical rule we arrived at: use MarpolBase for measured values, conditions, co-expression and curated literature in *M. polymorpha*; use TogoMCP to carry those results outward to structures, reactions, compounds, orthologs in other species and sequence archives; and bridge the two with KO, GO and compound identifiers rather than with gene identifiers.

Side project: DFAST-Agent for DDBJ submission

DFAST-Agent is an experimental prototype implementation of an AI-agent-based assistant for preparing DDBJ submission files. It was built during the hackathon to test whether the approach works at all; it is not a released tool. The submission procedure and the format converters are exposed to the agent as an MCP server, skills and tools. The agent reads and writes files on **Kura**, the authenticated object storage provided for DDBJ users, and collects the metadata a submission requires through dialogue with the user.

Prokaryotes. Using the DFAST API for prokaryotic genomes (Tanizawa et al., 2018), files held on Kura were converted into DDBJ submission files. Interactive preparation of a complete submission file from the terminal succeeded in the prototype.

Eukaryotes. A converter from GFF to DDBJ submission format was added. Terminal-based dialogue with the agent successfully collected the required metadata and produced the submission files. For users who do not work with an AI agent, we additionally built a web interface backed by a local LLM running on the NIG supercomputer, operating on files in Kura. Tuning the local-LLM agent proved difficult and the file conversion did not succeed in that configuration, but the work produced usable knowledge about operating against Kura. Development continues.

Side project: MARKit, a marker submission tool

Background

A submission of barcode sequences such as COI, 16S or ITS often contains hundreds to thousands of entries, yet the annotation is left to the submitter by hand. Coordinate shifts, CDS features that do not translate, and disagreement between the declared organism name and the sequence itself are sometimes found only after registration. **MARKit** (Marker Annotation and Registration Kit) builds DDBJ MSS-format submission files mechanically from a FASTA file plus minimal metadata, and inspects them before submission.

The pipeline is: marker assignment → region determination → structural check → organism-name reconciliation → submission-file generation → report.

Features

- **Markers are determined, not declared.** Every model (HMM / CM) is applied and the marker is decided per entry, so a submission containing mixed markers is processed separately per marker and merged into a single submission file at the end. 22 markers are supported.
- **The genetic code is chosen by taxon.** Mitochondrial code tables differ between taxa, so the taxon is looked up from the declared organism name before reading frames are evaluated.
- **Unregistrable shapes are avoided at generation time** — a reading frame that does not translate is not emitted as a CDS, and minus-strand hits are normalised to the plus strand with the coordinates moved accordingly.
- **Organism-name reconciliation.** Sequences are compared against a reference database by distance and adjudicated with per-family calibrated thresholds. Candidates are proposed even when no organism name is declared, which helps to settle the species assignment.
- **Metadata can come later.** A FASTA file alone is enough to run the analysis; the submission files can be built once the metadata is available.
- **Per-entry decisions after review.** Looking at the analysis, the submitter can withdraw an entry from the submission, or register it as `misc_feature` / `misc_RNA` instead of `CDS` / `rRNA`.
- **Validated against real submissions.** Generated output is compared against the .ann files of 91 actual submissions (about 10,000 entries), and this check is re-run whenever a feature is added.
- **Distributed as a container** carrying the runtime, the reference data and the code (Apptainer 657 MB / OCI 4.7 GB, published as `nigyt/markit:latest`).

Web interface

To widen the user base, MARKit was embedded in the existing DDBJ Gene/Genome Submission (GGS) tool as a web interface (Figures 3 and 4). MARKit itself is invoked as a container, so the analysis logic is identical to the command-line version. The interface has five screens — INPUT, ANALYSIS, BIOLOGICAL SOURCE, COMMON, BUILD & VALIDATE — and several FASTA files can be analysed with different marker settings in one job. Screens are bilingual (Japanese and English), and all uploads are validated server-side.

MARKit: Marker Annotation and Registration Kit

Submits barcode sequences of phylogenetic markers (COI / ITS / 16S and others).
It decides each entry's marker and feature boundaries from the sequence itself, checks the declared organism name against a reference database, and writes the DDBJ registration files.
Several FASTA files can be submitted with their own marker settings and are merged into a single submission.

You are not logged in. The job is deleted after 24 hours and does not appear in your history. Keep this page's URL (it carries the job_id) to come back to the same job, and download the registration files you build. Submitting requires a login.

1. INPUT 2. ANALYSIS 3. BIOLOGICAL SOURCE 4. COMMON 5. BUILD & VALIDATE

One set = one FASTA plus one marker setting. Several sets are merged into a single submission (one .ann / .fasta at the end). Entry names must therefore be unique across all sets; duplicates are rejected here.

Set 1 20251225test_COXI.fasta

Marker
auto

+ Add a set

Metadata (optional, one per job)
You can hand over common.json and the sample metadata TSV here as well. The first column of the TSV must match the ID in the FASTA header (everything up to the first space). When both files are present and complete, the analysis and the registration files are built in one go. You can also supply them later on the COMMON and BIOLOGICAL SOURCE tabs.

Choose common.json Choose sample TSV

Run analysis

Figure 3: The MARKit web interface, INPUT screen. Each FASTA file is paired with a marker setting (auto lets MARKit decide), and common.json and the sample metadata TSV can be supplied here or later.

11 entries

BLOCKED 0 OK 0 REVIEW 11

Markers: COI 11

BLOCKED entries are left out of the registration files (they stay in this table). REVIEW entries need a human check.

Looking at the analysis, you can withdraw an entry from the submission or register it as misc_feature / misc_RNA instead of CDS / rRNA. MARKit rebuilds the registration files; it also decides whether a demoted entry becomes misc_feature or misc_RNA. A note goes into /note on the demoted feature.

BLOCKED REVIEW OK

entry	marker	status	organism	verdict	action	location	codon_start	query_coverage	alerts	notes
AB491414_Echinococcus	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>789	1	0.99		three_prime_partial
AB688152_Phytophthora	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>672	1	0.99		three_prime_partial
AB948168_Skeletonema	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>997	2	0.99		three_prime_partial
HQ860420_Bostaurus	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>658	2	0.99		three_prime_partial
KJ918252_3-terminal-partial	COI	REVIEW		UNVERIFIABLE	REVIEW	25..>1490	1	0.90		three_prime_partial;start_r
KJ918330_start-end_with_UTR	COI	REVIEW		UNVERIFIABLE	REVIEW	25..1572	1	0.57		start_refined_by_hmm;mod
LC798357_forward_CDS	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>658	2	0.99		three_prime_partial
LC798358_complement_CDS	COI	REVIEW		UNVERIFIABLE	REVIEW	complement(<1..>658)	2	0.99	revcompl	three_prime_partial
LC798359_misc_feature	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>349	2	0.51	cdsstop	orf_selected_by_hmm;orf_t
LC798360_forward_CDS	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..>658	2	0.99		three_prime_partial
MW361241_3UTR	COI	REVIEW		UNVERIFIABLE	REVIEW	<1..1532	3	0.51		model_exhausted

Save registration choices Chosen: 0 not registered / 0 demoted

Figure 4: The MARKit web interface, per-entry analysis table. Each entry carries its assigned marker, status, feature location, codon start, query coverage and alerts; from here an entry can be withdrawn from the submission or demoted to misc_feature / misc_RNA.

Status

The command-line version is usable in production, and all five screens of the web version work end to end.

Discussion

Publishing MarpolBase as RDF made it possible to link a species-specific plant genome resource to external resources, and registering the endpoint with TogoMCP and RDF Portal contributed a plant genome resource to the shared Japanese infrastructure. The most transferable lesson is the one in Table 2 and the section that follows it: for a resource covering a species that the major cross-reference hubs do not index, the integration point is not the gene identifier but the shared vocabularies the resource chooses to assign — in our case, above all, its own KO assignments.

On the DDBJ side, both DFAST-Agent and MARKit reduce manual work in submission preparation, but they are at different stages and rest on different assumptions: DFAST-Agent is still a prototype and collects what it needs through dialogue, which makes it dependent on a capable agent, while MARKit derives what it can from the sequence itself and asks the submitter only where a decision is genuinely required. The local-LLM experiment showed that the dialogue-based approach is currently hard to reproduce without a frontier model, which is an argument for keeping the deterministic path available alongside it.

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Author contributions

Yasuhiro Tanizawa: Conceptualization, Software, Writing – original draft; Takatomo Fujisawa: Software, Writing – review & editing.

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